Boda Meetkumar Mansukhbhai

Master of Technology
Department of Computer Science and Engineering
Indian Institute of Technology Kanpur

MeetBoda
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Education

Year	Degree/Certificate	Institute	CPI/%
2025–Present	M.Tech (Computer Science & Engineering)	Indian Institute of Technology, Kanpur	**9.5**/10
2021–2025	B.Tech (Computer Engineering)	Dharmsinh Desai University, Nadiad	9.68/10
2021	CBSE (Class XII)	Sainik School Balachadi, Jamnagar	95.4%
2019	CBSE(Class X)	Sainik School Balachadi, Jamnagar	95.0%

Research Experience

- **Scaling Laws for LLMs for Indic Languages** (MTech Thesis, Guide: Dr. Arnab Bhattacharya) *(Ongoing)*
  - Using 106M documents from the **Hindi dataset of IndicCorpv2**, trained a custom tokenizer with a **50k vocabulary size** achieving a **fertility of 1.16** and then along with Bangla dataset trained another tokenizer with 70k vocab size.
  - Developed the pretraining pipeline and pretrained multiple **ELECTRA-Small models** with data fractions **ranging from 1%, 5%, 10%, 20%, 40%, 60%, 80%, to 100%** for Hindi and Bilingual Data.
  - Built a benchmark evaluation suite utilizing IndicGLUE and IndicXTREME spanning 4 core tasks: **NER, WSTP, MASSIVE-Intent, and MASSIVE-Slot**. Benchmark suite was ran on all the pretrained models.
  - Based on experiments carried out till date, trends indicates that while scaling pretraining data consistently reduces loss, it **does not directly translate into downstream task performance**.
  - Research Areas: Large Language Models, Natural Language Processing, Scaling Laws

Projects

- **Recall-Optimized Planetary Crater Detection** (Course Project: CS776) *(Mar – Apr 2026)*
  - Technologies : Swin Transformer, UNet++, Streamlit
  - Engineered an end-to-end deep learning pipeline for Martian crater detection on THEMIS Day IR imagery ($\sim$100m/px), utilizing a recall-critical objective to eliminate false negatives in planetary cataloguing.
  - Architected **SwinCraterNet** (36.1M params) with a Swin Transformer backbone, ASPP bottleneck, Circular Geometry Gate, and dilated residual decoder, achieving SOTA **0.8156 recall** (surpassing published baselines).
  - Designed **RDT-UNet++**, an edge-efficient **4.62M-parameter** model combining multi-scale dilated convolutions ($\{1, 2, 4\}$), Circular Transformers, and CBAM gates, delivering **0.7574 recall** with $> 20\times$ parameter reduction.
  - Developed a physics-aware **copy-paste augmentation** framework with 65% small-crater biased sampling, collision-free placement, and Gaussian blending to resolve extreme scale imbalance and small-crater recall deficits.
  - Built a post-processing pipeline using multi-radius circular template matching to generate structured CSV catalogues (x, y, r), deployed via an interactive Streamlit inference application.
- **3D Stencil Computation and Isovalue Edge Counting** (Course Project: CS633) *(Mar – Apr 2026)*
  - Technologies : C++, MPI
  - Implemented a **distributed 3D MPI stencil computation** with domain decomposition, performing configurable d-point neighbor averaging and **isovalue counting** across time steps.
  - Optimized inter-process communication using **non-blocking 6-face halo exchanges with Isend/Irecv**, overlapping communication with computation to reduce synchronization overhead.
  - Deployed and benchmarked the implementation on the PARAM Rudra cluster using SLURM, evaluating scalability across 32–96 processes and varying 3D data dimensions.
- **Inference of Spatial Metacells from Single-Cell and Spatial Data** (Course Project: CS690) *(Oct – Nov 2025)*
  - Technologies : ENVI, MetaQ, COVET, UMAP
  - Formulated a **multimodal spatial metacell framework** integrating mouse motor cortex scRNA-seq (**71.2k cells** $\times$ 30.6k genes) with MERFISH spatial transcriptomics (**276.5k spots** $\times$ 254 genes) to preserve cellular niche microenvironments.
  - Developed a **post-hoc latent splitting pipeline** using ENVI's Conditional VAE co-embeddings and k-Means clustering on transcriptomic metacells, boosting **average compactness to 0.804** (surpassing the 0.775 scRNA baseline).
  - Modeled spatial niche interactions by computing shifted gene-gene covariance matrices (**COVET**) across k-NN neighborhoods and extracting **25-dim spectral eigenvalue signatures** via SVD to capture localized gene program variance.
  - Benchmarked 4 integration strategies**Latent Splitting, Spectral Eigenvalue Augmentation, Shared Latent Space, and AOT-Leiden Graph Clustering**evaluating compactness and separation trade-offs via UMAP projections.
- **Automated Subjective Answer Grader** (Course Project: CS787) *(Sep – Nov 2025)*
  - Technologies : DeBERTa-v3-large, BigBird-RoBERTa-large, Streamlit
  - Architected a **hybrid multi-model grading framework** decomposing subjective grading into LLM rubric genera-

tion, multi-engine answer segmentation, and evidence-grounded scoring to eliminate length bias and hallucinations.
  - Fine-tuned **DeBERTa-v3-large** (435M params) on SQuAD 2.0, achieving **85.81% Exact Match (EM)** and **89.06% F1-score**, enabling zero-hallucination, token-level span extraction for CPU-efficient inference.
  - Benchmarked 3 selectable segmentation engines**GPT-OSS 120B** (generative multi-span synthesis), **DeBERTa-v3 QA** (extractive spans), and **all-mpnet-base-v2** (cosine semantic similarity)to balance latency and recall.
  - Investigated long-context QA by fine-tuning **BigBird-RoBERTa-large** (sparse attention, 4,096 tokens) on MASH-QA and HotpotQA, analyzing multi-hop reasoning shifts and score collapse in quantized Qwen-2.5 3B.
  - Built an interactive **Teacher-in-the-Loop Streamlit UI** featuring dynamic verbatim span highlighting, real-time rubric refinement, and transparent score overriding.
- **State-wise Classification of Indian Traditional Garments** (Self Project) *(Feb – Mar 2025)*
  - Technologies : ViT, EfficientNet, CNN, Selenium
  - Curated and cleaned a dataset of **3,748 verified images** across 8 Indian states using **Selenium** web scrapers and manual verification, applying data augmentation to mitigate class imbalance.
  - Fine-tuned a **Vision Transformer (ViT)** to capture long-range dependencies across intricate textile motifs and draping styles, achieving **94.9% test accuracy** and **0.95 macro F1-score**.
  - Designed and benchmarked 3 deep learning modelsa **Custom 4-layer CNN** (66.7%), **EfficientNet-B0** (89.5%), and **ViT** (94.9%)evaluating efficiency versus accuracy trade-offs.
  - Conducted class-wise confusion matrix diagnostics across 571 test images to analyze cross-regional visual overlaps and feature saliency (e.g., traditional ornaments and fabric textures).
- **KrushiMitr** (Self Project) *(Dec 2023 – Apr 2024)*
  - Technologies : React.js, Node.js, Express.js, MongoDB, Flask, TensorFlow, OpenCV
  - Engineered a full-stack agricultural platform using the **MERN stack** and **Flask**, enabling farmers to consult certified experts and access localized farming resources and community of fellow farmers.
  - Integrated a computer vision classification model using **VGG19** to diagnose 25+ crop leaf diseases from images, coupled with **OpenAI API** to generate actionable remediation steps.
  - Developed role-based portals (Farmer & Expert) supporting interactive Q&A forums with voting, real-time weather tracking, multi-lingual support and access to historical agricultural data.
  - Designed modular **RESTful APIs** and **MongoDB** schemas to handle image uploads, query triage, and state/district-wise crop data management.
- **Attendance Management System** (Self Project) *(Sep – Oct 2023)*
  - Technologies : .NET Framework, C#, ASP.NET, SQL Server
  - Developed a full-stack web application using **.NET Framework** and **SQL Server**, featuring role-based access control (RBAC) for independent Teacher and Student workflows.
  - Engineered subject-wise attendance logging and semester-wise historical reporting dashboards, automating record retrieval and attendance calculations.
  - Streamlined academic administrative operations by digitizing manual paper-based workflows for course allocation and profile tracking.
- **Student Classroom** (Self Project) *(Jan – Mar 2023)*
  - Technologies : Python, Django, JavaScript, HTML5/CSS3, Bootstrap, SQLite, Selenium WebDriver, PyTest
  - Developed a web-based learning platform using **Django** and **Bootstrap**, facilitating dynamic class-code enrollment, participant management, and centralized course material sharing.
  - Implemented **Role-Based Access Control (RBAC)** with automated submission verification (on-time vs. late tracking), deadline enforcement, and an interactive evaluation/grading portal.
  - Designed a collaborative doubt-resolution forum supporting student queries and teacher-verified answer badges (Verified/Unverified) for accurate peer learning.
  - Built an end-to-end testing suite using **Selenium** and **PyTest** following the **Page Object Model (POM)** architecture, enabling automated cross-browser validation and report generation.
- **College360: 3D Virtual Campus Tour** (Self Project) *(Feb 2023)*
  - Technologies : JavaScript, Three.js, Panolens.js, Flask
  - Built a web-based 3D virtual campus tour for DDU Nadiad using **Panolens.js** and **Three.js**, providing interactive 360° panoramic navigation similar to Google Street View.
  - **Implemented an interconnected hotspot navigation graph** across campus landmarks to enable seamless spatial scene transitions and point-of-interest inspection.
  - Optimized WebGL rendering and texture loading for low-latency, 60 FPS performance on both mobile and desktop browsers.

Academic Achievements

- **Reliance Foundation Postgraduate Scholar** *(2025–2027)*
- **Gold Medalist in B.Tech, Computer Engineering (Class of 2025)** – Dharmsinh Desai University *(2025)*
- **School Topper in Class XII** – Sainik School Balachadi *(2021)*

- **Best Open Innovation Award** – DUHacks 2.0 (Project: College360) *(2023)*
- **Best All Rounder Cadet of the School** – Sainik School Balachadi *(2019–2020)*
- **Cleared UPSC NDA/NA 2020(II)** in first attempt. (**SSB Recommended**) *(2021)*

Technical Skills

- **Languages:** Python, C/C++, SQL, JavaScript, C#, Java
- **ML/AI:** PyTorch, Tensorflow, Scikit-Learn, Transformer, LLMs, Computer Vision
- **Data Science/Libraries :** NumPy, Pandas, Matplotlib, Seaborn, SciPy
- **Generative AI:** LangChain, LangGraph, MCP, RAG, AI Agents, LLM Fine tuning
- **Frameworks & Web:** ReactJS, NodeJS, ExpressJS, MongoDB, Django, .NET, HTML, CSS, Flask, Selenium

Relevant Courses

- **MTech Courses:** Introduction to Machine Learning, Deep Learning for Computer Vision, Probabilistic Machine Learning, Computational Genomics, Generative Artificial Intelligence, Parallel Computing
- **BTech Courses:** Operating Systems, Computer Organization and Architecture, Database Management Systems, Software Engineering Principles, Data Structure and Algorithm, Design and Analysis of Algorithms, Digital Logics

Positions Of Responsibility

- **Student Guide** – CMHW, IIT Kanpur *(June 2026 – Present)*
  - Mentored group of 6 Y26 PG students in matters related to orientation, academics and admission process.
- **Teaching Assistant** – ESC111/112M & CS771, IIT Kanpur *(Aug 2025 – Present)*
  - Helped professor in conducting lab sessions, examinations and evaluating the answer sheets.

Extra-Curricular Activities

- **National Cadet Corps (NCC 2015–2021):** Completed 6 years of military discipline and leadership training, earning **Certificate 'A'** and **Certificate 'B'** with top merit and also attended 3 CATC Camps. *(2015–2021)*
- **Quizzing:** Represented the school in district/state/national quizzes including **2nd Position in All India Inter School Online Quiz** conducted by Sainik School UP, Lucknow in November 2020. *(2017–2021)*
- **Sports:** Active player in **Cricket** and **Volleyball** in inter-house events. Also an avid **long distance runner**, with participation in 3 Marathons and various other races. *(2017– Present)*